Rover Acceleration Characterization

Measure rover motion and determine whether acceleration is approximately constant during startup.

Intermediate	1 class period	Skill Builder

Engineering Context

A mission-planning team needs a simple model of rover startup motion to estimate stopping and maneuvering distance.

What You Will Practice

- Collect position-time data
- Calculate velocity and acceleration
- Interpret motion graphs
- Judge whether a model is appropriate

What You Need

- Rover
- Straight measured track
- Video camera or motion sensor
- Spreadsheet
- Engineering notebook

Student Instructions

1. Mark a straight test track and choose a consistent rover command.
2. Collect position versus time using video frames or a motion sensor.
3. Create a position-time graph.
4. Calculate interval velocities and create a velocity-time graph.
5. Estimate acceleration from the graph.
6. Repeat the trial two more times.
7. Compare acceleration values and decide whether a constant-acceleration model is reasonable.

Engineering Requirements

- At least three trials
- Consistent battery and command settings
- Show calculation method for velocity and acceleration
- Use graphs with labeled axes and units
- Model conclusion must include evidence

Constraints & Safety

- Use a clear track
- Maintain immediate stop control
- Do not stand in front of the rover
- Secure loose cables and equipment

What You Submit

- Test method
- Position data
- Position-time and velocity-time graphs
- Acceleration estimate
- Model evaluation

Reflection Questions

1. Where was acceleration most constant?
2. How did sampling interval affect the graph?
3. What caused differences among trials?
4. How would payload mass change startup motion?

Engineering Tip Use consistent time intervals and identify the center time for interval velocity values.	**Common Mistake** Calculating one acceleration value from the entire run can hide changes during startup and steady motion.
Stretch Goal Repeat with an added payload and compare acceleration statistically.	**Career Connection** Vehicle dynamics and controls engineers characterize acceleration, braking, and traction for autonomous vehicles.